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Professor Sirazum Tisha

CMS 320

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Final Project Report

1. **Game Title:** Sweet Harvest Run

Authors: Caitlyn Lefebvre and Isabella Hamilton

GitHub Link: [caitlyn124/CMS320-Fall2025: Group Project](https://github.com/caitlyn124/CMS320-Fall2025-Group-Project)

2. **Introduction:**

a. **Background & Motivation**

We took inspiration from Caitlyn's game proposal for Assignment 2, but made some adjustments to simplify the concept and make it more suitable as a 2D game for beginner developers. Our goal is to create a fun, seasonal game that is enjoyable for players to experience.

b. **Target Users**

Although primarily designed for children aged 8 and up, *Sweet Harvest Run* is meant to be fun and engaging for players of all ages who enjoy lighthearted fantasy and adventure games.

c. **Context/Game World**

The game is set during the fall season in a large, winding corn maze. This seasonal theme creates a playful and festive atmosphere. Within the maze, players will encounter various obstacles and hidden pieces of candy corn scattered

throughout. The objective is to collect all the candy corn and find the exit before time runs out.

d. Rules of the Game

- Collect all the pieces of candy corn in each level. A counter at the top of the screen will show the player's progress.
- Escape each level before time runs out. A timer will be visible at the top of the screen.
- Players must avoid or overcome obstacles within the maze that may slow them down or block their path.
- Levels increase in difficulty as the game progresses, with more complex mazes and harder-to-find candy corn.

e. Controls

- Arrow keys: Move the Scarecrow in the respective direction.
- Space bar: Jump over obstacles.
- "C" key: Collect pieces of candy corn.

f. Characters & Plot

Main Character – The Scarecrow (Player):

The player controls a friendly Scarecrow whose mission is to collect candy corn and escape each level of the maze before time runs out.

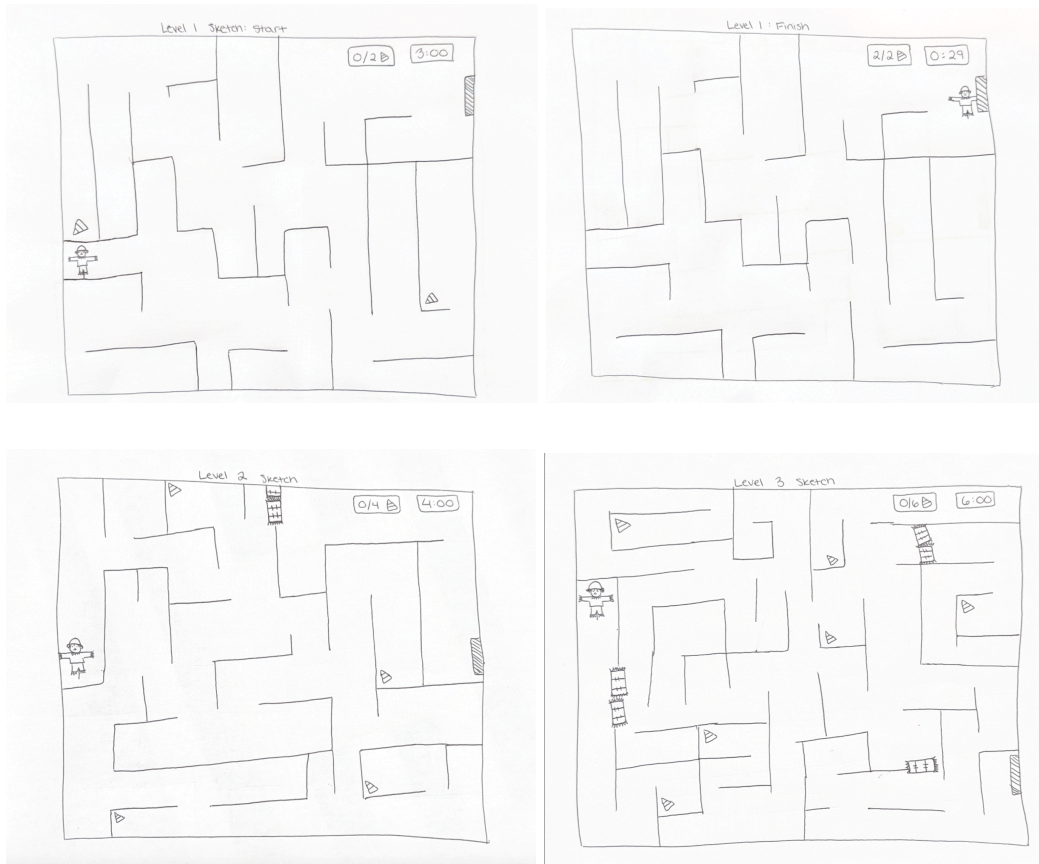
Antagonist – Time:

There is no physical enemy in the game; instead the challenge comes from racing against the clock.

Narrative Arc:

At the start of the game, a short dialogue introduces the player to their role as the Scarecrow and the challenge that lies ahead: collect all the candy corn and escape the corn maze. As the player progresses through each level, the maze becomes more difficult. The game concludes when the Scarecrow collects all candy corn and escapes the final level, rewarding them with the chance to enjoy the rest of fall.

g. Rough Sketch



h. Distribution

- Caitlyn will serve as the story writer and programmer, responsible for developing the narrative elements (plot, dialogue, conflict, etc.) and assisting in programming tasks.
- Isabella will be the artist and programmer, creating all the visual assets for the game and also contributing to the programming and implementation.

3. Game Development

Caitlyn: 10/19/2025

- Created new 2D project called “CMS320-Fall2025”
- Created folders called “Sprites” and “Scripts”
- Added new scene called “HomePage”
- Created a canvas object called “Canvas”
- Added text object called “Title”
- Added two buttons called “RulesButton” and “PlayButton”
- Created a new C# script called “MainMenuController” and attached it to the “Canvas” object
- Attached the “Canvas” object to each button

Isabella: 10/19/2025

- Created the game outline/rough drafts
- Created sprite artwork for haybale
- Created sprite artwork for candycorn

Caitlyn: 10/24/2025

- Created new scene called “RulesPage”

- Created a canvas object called “Canvas”
- Added text object called “RulesText”
- Added button called “BackButton”
- Created a new C# script called “RulesPageController” and attached it to appropriate buttons
- Created new scene called “Level1”
- Updated “MainMenuController” script

Isabella Hamilton: 10/25/2025

- Created artwork for Scarecrow Animation: Walking forward
- Created artwork for Scarecrow Animation: Walking backwards
- Created artwork for Scarecrow Animation: Walking left
- Created artwork for Scarecrow Animation: Walking right
- Created artwork for Scarecrow Animation: Jumping forward
- Created artwork for Scarecrow Animation: Jumping backwards
- Created artwork for Scarecrow Animation: Jumping left
- Created artwork for Scarecrow Animation: Jumping right
- Created artwork for Scarecrow Animation: Idle forward
- Created artwork for Scarecrow Animation: Idle backwards
- Created artwork for Scarecrow Animation: Idle left
- Created artwork for Scarecrow Animation: Idle right

Caitlyn: 10/26/2025

- Created a new folder called “Images”
- Added scarecrow sprites to “Images” folder

- Created an animator controller called “ScarecrowAnimator”
- Added animator component to “Scarecrow” object and dragged “ScarecrowAnimator” into it
- Created a new folder called “Animation”
- Created the following animations with the appropriate scarecrow images: “IdleUp”, “IdleDown”, “IdleLeft”, “IdleRight”, “WalkUp”, “WalkDown”, “WalkLeft”, “WalkRight”, “JumpUp”, “JumpDown”, “JumpLeft”, “JumpRight”
- Opened the animator window and added the following parameters: “MoveX” (float), “MoveY” (float), “IsMoving” (boolean), “Jump” (trigger)
- Created a new C# script called “ScarecrowController” and attached it to the “Scarecrow” game object

Caitlyn: 10/30/2025

- In the animator window, created three new blend trees called “IdleBlendTree”, “MoveBlendTree”, and “JumpBlendTree”
- Set each to have parameter x as “MoveX” and parameter y as “MoveY”
- Added 4 motion fields to each for each direction and assigned the corresponding animations
- Created transitions between all three blend trees with corresponding “IsMoving” and “Jump” conditions
- Updated “ScarecrowController” script

Isabella: 10/30/25

- Created Sprite for cornstalk

Caitlyn: 10/31/2025

- Added cornstalk sprite to the “Sprites” folder
- Added “BoxCollider2D” and “Rigidbody2D” components to Cornstalk game object
- Resized Cornstalk game object, duplicated, and made corn maze
- Updated “ScarecrowController” script

Caitlyn: 11/1/2025

- Created empty object called “Boundaries”
- Within “Boundaries”, created four empty game objects: “TopWall”, “BottomWall”, “LeftWall”, and “RightWall”
- Added “BoxCollider2D” to each and adjusted their positions and sizes to form a barrier around the camera’s field of view

Isabella: 11/3/2025

- Created background for home page
- Created artwork for play button
- Created artwork for rules button
- Created artwork for back button
- Created background for rules page
- Created grass background artwork

Caitlyn: 11/5/2025

- Added/moved cornstalk sprites
- Added background image to “HomePage” and “RulesPage” scenes
- Added button images for “RulesButton”, “PlayButton”, and “BackButton”

Isabella: 11/13/25

- Created artwork for open door sprite
- Created artwork for closed door sprite
- Created artwork for restart button
- Created artwork for home button

Caitlyn: 11/15/2025

- Fixed the scaling of the “HomePage” and “RulesPage” scenes
- Imported the Candy Corn sprite
- Created a Candy Corn prefab
- Created the “CandyManager” script
- Created the “CandyCorn” script
- Added “Player” tag to scarecrow
- Added “CandyCorn” prefabs to the scene
- Imported Grass Background image
- Positioned Grass Background image in scene
- Fixed “CandyManager” script
- Imported Open Door and Closed Door sprites
- Added Closed Door sprite to scene
- Created “DoorController” script
- Connected CandyManager to the door

Caitlyn: 11/16/2025

- Imported hay bale sprite
- Created a Haybale prefab
- Updated “ScarecrowController” script

- Added “Haybale” prefabs to the scene
- Added “CameraFollow” script
- Made the “Scarecrow” sprite the Main Camera’s target
- Changed the size of the camera
- Changed the size of the Grass Background image
- Added more Cornstalk sprites

Isabella: 11/18/25

- Added counting down timer for each level
 - Added level timer script
- Added flashing color to timer when it reaches a low time
- Added background music that starts when the game starts and plays continuously
 - Added background music script
- Added sound effects for collecting candycorn
 - Added changes to candycorn script
- Added sound effects for the door opening when x number of candy corn are collected
- Added a title on the top of each level
- Created a tutorial level to show the players how to play the game.
 - Added tutorial trigger script
 - Created a simple cornstalk maze
 - Added two candy corns to be collected
 - Added a hay bale to jump over

- Includes text popups on screen when the player enters a certain zone that gives direction on what to do.
- Created a screen popup when the player runs out of time.
 - Added options to restart the level or return to the home page on this popup
 - Done through buttons for restart and home
 - Added “Time’s Up!” text on popup
 - Created Game over Buttons script
- Created the artwork for the final “You Win” page.
- Added the final “You Win” page.
 - Added a button that brings you back to the home page.

Caitlyn 12/1/2025

- Fixed “CandyCorn” script
- Cleaned up “Tutorial” scene
- Added “Level3,” “Level4,” “Level5” scenes
- Finalized time constraints for “Level1,” “Level2,” “Level3,” “Level4,” and “Level5” scenes

Isabella: 12/2/25

- Added the start up page with our names and graduation year
- Attached startup script that allows the player to press the space bar and be brought to the home page to begin the game
- Made sure the level orders were correct

4. Playtest Report (11/12/2025)

Three things to improve:

1. Adding a “preview” level before the first level so players can learn the controls before starting the actual game.
2. Adding background music to make the game more immersive.
3. Adding another level of difficulty to the game (whether that’s an enemy to run away from or reducing the visibility of the player). Even though this game is made for children, the difficulty isn’t hard, and it could get boring after a while.

5. **Final Game**

https://drive.google.com/file/d/1c5fd_-ld4UXY1OxR4EmstYiW2ulIyHvW/view?usp=sharing

6. **Discussion & Future Work**

Working on this project gave us a new appreciation for games because we finally got to see everything from the developer’s side. It was fun, but also eye-opening to realize how much time and effort goes into even the simplest mechanics. Throughout the process we used YouTube tutorials and in-class examples to help us build different parts of the game and figure things out as we went. Based on feedback, some features just weren’t realistic to add within the short timeframe, but we still tried to make the game more challenging like people suggested. Looking ahead, we talked early on about creating different themed versions, like Christmas or Easter, which would be a fun direction to expand the game in the future.